COMMON COURSE ASSIGNMENT

CSA65 – GENERATIVE AI AND LARGE LANGUAGE MODELS

Enterprise Knowledge Assistant Using Retrieval-Augmented Generation and AI Agents

CO2, CO3 & CO4 | *Units II, III & IV*

Assignment Information	Details
Department	Computer Science and Engineering
Programme	B.E. / B.Tech – CSE
Course Code & Name	CSA65 – Generative AI and Large Language Models
Academic Year / Batch	2026 – 2027
Faculty Name	Dr. K. Malathi
Assignment Title	Enterprise Knowledge Assistant Using Retrieval-Augmented Generation and AI Agents
Date of Issue	01-09-2026
Date of Submission	03-09-2026
Maximum Marks	100
Course Outcomes	CO2, CO3, CO4
Bloom's Taxonomy	L3 – Apply, L4 – Analyze, L5 – Evaluate
SDG Mapping	SDG 9 – Industry, Innovation and Infrastructure

1. Problem Understanding and Formulation

KnowledgeDesk is an internal assistant for a 4,000-employee engineering services company. Employees should be able to ask natural-language questions about HR policies, quality SOPs, product manuals, project reports and compliance documents instead of searching a shared drive. The corpus is about 12,000 documents, averaging eight pages, with PDF, DOCX and scanned-image content. The main challenge is to return grounded, cited answers while respecting access control, latency, cost, storage and reliability requirements.

- Prompting: maintain consistent grounded answers inside an 8,192-token context window.
- Retrieval: reach Recall@5 ≥ 0.85 over a large and growing document collection.
- Agents: handle multi-step requests through bounded tool use.
- Multimodal: preserve information in scans, tables and screenshots.
- Governance: enforce RBAC, protect PII, log sources and provide human escalation.

2. Application of Course Knowledge – CO2, CO3, CO4

Part I – Prompt Engineering & LLM Behaviour

Zero-shot

```
Answer clearly.
Question: What is the annual leave entitlement under the current KnowledgeDesk HR policy?
```

Few-shot

```
Use evidence and cite document/page. Example: Q: Who approves leave? A: Reporting manager [HR_Leave_Current.pdf, p.1].
Now answer: What is the annual leave entitlement under the current KnowledgeDesk HR policy?
```

Chain-of-thought controlled

```
Reason internally but do not reveal private chain-of-thought. Give only the conclusion and evidence citation.
Question: What is the annual leave entitlement under the current KnowledgeDesk HR policy?
```

Role-based grounded

```
SYSTEM: You are KnowledgeDesk. Answer only from evidence; cite every factual claim as [document, p.X]; abstain when evidence is insufficient.
TASK: What is the annual leave entitlement under the current KnowledgeDesk HR policy?
EVIDENCE: [HR_Leave_Current.pdf, p.1] Annual leave entitlement is 24 working days per calendar year.
```

Recommended strategy: role-based grounded prompting combined with a few-shot example for citation and refusal behaviour. Temperature should normally be 0–0.2 for policy/compliance questions, with an explicit evidence-only rule and a controlled answer length.

Context-window derivation

```
400 + 600 + 512k + 600 ≤ 8192
512k ≤ 6592
k ≤ 12.875
Maximum k = 12
```

Monthly API cost

```
Input/query = 400 + 600 + (12×512) = 7,144 tokens
Input = 7,144/1,000 × ₹0.18 = ₹1.28592
Output = 600/1,000 × ₹0.55 = ₹0.33
20,000 queries = 20,000 × ₹1.61592 = ₹32,318.40
```

The calculated cost is ₹7,681.60 below the ₹40,000 monthly budget.

Part II – Retrieval-Augmented Generation & AI Agents

```
Total tokens = 12,000 × 8 × 500 = 48,000,000
Tokens/document ≈ 4,000
Stride = 512 − 50 = 462
Chunks/document ≈ ceil((4,000−50)/462) = 9
Estimated chunks = 12,000 × 9 = 108,000
```

Using 768-dimensional float32 vectors: 108,000 × 768 × 4 = 331,776,000 bytes, approximately 0.332 GB decimal (0.309 GiB). This is comfortably below 100 GB before adding metadata, index overhead and source documents.

Retrieval evaluation design

Configuration	Chunk/retrieval	Precision@5	Recall@5	MRR
A	512, dense, top-5	0.78	0.83	0.75
B	512, hybrid dense + BM25, top-5	0.86	0.92	0.88
C	384, hybrid + rerank, top-5	0.90	0.94	0.91

The above comparison is a representative prototype table for the report. It should be replaced with measurements from the final corpus before submission. The ZIP includes a 12-question evaluation CSV and evaluate.py to compute Precision@5, Recall@5 and MRR.

AI agent

Tool	Purpose
Vector retriever	Retrieve authorised evidence
Document metadata lookup	Find document/version/department information
Calculator	Perform deterministic arithmetic
Ticket creation API	Create a follow-up ticket when explicitly requested

Routing rule: ordinary factual questions use RAG; queries containing compare, calculate, find-document or ticket intent are escalated to the agent. The prototype uses a bounded tool loop so actions remain auditable and latency remains predictable.

Part III – Multimodal GenAI & Intelligent Applications

Approach	Strength	Limitation	Decision
OCR → text embedding	Simple and inexpensive	Can lose layout and visual relationships	Baseline
Vision-language captioning → text embedding	Captures richer visual meaning	Extra inference cost and caption errors	Fallback
Native multimodal embeddings	Direct text/image similarity	More deployment complexity	Long-term recommendation

Latency analysis

```
Generation = 250/45 × 1000 ≈ 5,556 ms
```

Pipeline	Total latency	Meets ≤3 s?
No reranking	40 + 25 + 5,556 = 5,621 ms	No
Rerank top-20	40 + 25 + 2,400 + 5,556 = 8,021 ms	No
Top-5 + 150 tok/s generator	40 + 25 + 600 + 1,667 = 2,332 ms	Yes

The original 45-token/second generation rate cannot satisfy the 3-second requirement for a 250-token answer. The final design therefore recommends a faster generator, controlled answer length and top-5 reranking. Caching and early exit can further reduce latency.

3. Solution / Design / Methodology

3.1 Integrated architecture

```
Employee → Authentication/RBAC → Query Router
                            ├─ factual → Hybrid Dense + BM25 → Reranker → Grounded LLM → Citations
                            └─ multi-step → Agent → Retriever / Metadata / Calculator / Ticket API → Answer
Documents → PDF/DOCX/TXT/OCR → Page-aware chunks → 768-D embeddings → FAISS
```

3.2 Prompting design

```
You are KnowledgeDesk, an internal enterprise knowledge assistant.
Answer only from retrieved evidence.
Cite every factual statement as [document, p.X].
If evidence is missing, conflicting or below threshold, abstain.
Never invent dates, numbers, names or procedures.
Respect the user's role and authorised departments.
Keep the response concise and do not reveal private chain-of-thought.
```

3.3 Retrieval design

- 512-token chunks with 50-token overlap.
- all-mpnet-base-v2 embeddings (768-D).
- FAISS inner-product search on normalized vectors.
- Hybrid dense + BM25 retrieval.
- Top-20 candidates, then top-5 reranking.
- RBAC filtering before generation.
- Confidence threshold with explicit abstention.

3.4 Agent and application design

- Plain RAG for normal one-step questions.
- Bounded agent for multi-step/tool requests.
- Streamlit demonstration interface with visible sources.
- FastAPI is the recommended production backend for scaling.
- Audit logs store role, query, retrieved sources, model version, latency and refusal state.

4. Use of Modern Tools

Tool	Role
Python	Core application and evaluation
Sentence Transformers	768-D embeddings
FAISS	Vector similarity search
BM25	Lexical retrieval
Cross-Encoder	Candidate reranking
Ollama	Local LLM inference
Streamlit	Interactive interface
PyMuPDF / python-docx	PDF/DOCX parsing
Tesseract (optional)	OCR for scanned PDFs

5. Results and Validation

Validation	Requirement	Finding
Context window	≤8,192 tokens	k=12
Inference cost	≤₹40,000/month	₹32,318.40 estimate
Raw vectors	≤100 GB	≈0.332 GB decimal
Recall@5	≥0.85	Representative hybrid/rerank: 0.92–0.94
Latency	≤3 s	Needs faster generation than 45 tok/s
RBAC	No cross-department access	Department filter before generation
Abstention	At least 3 out-of-corpus tests	Threshold implemented; validate on final corpus

The report deliberately labels representative retrieval values as such. The implementation should be rerun with the actual final document set and the measured values inserted before academic submission.

6. Analysis and Engineering Decision

Hybrid retrieval is selected because dense embeddings capture semantic similarity while BM25 protects exact policy terminology and identifiers. Top-5 reranking gives a useful quality improvement without the 2.4-second overhead of reranking 20 candidates. The agent is used only when the request actually needs tools, avoiding unnecessary latency and complexity.

- Use local embeddings and local LLM inference to reduce recurring hosted API usage.
- Use caching for repeated policy questions and embeddings for stable documents.
- Apply RBAC before the generator sees retrieved text.
- Retain document version and page metadata for auditable citations.
- Use human escalation for sensitive or uncertain cases.

7. Broader Considerations

Area	Design response
Reliability & Safety	Evidence threshold, mandatory citations, abstention, logging and human escalation.
Sustainability / Economics	Local models, caching, limited reranking and short outputs.
Accessibility & Society	Natural-language interface, concise answers and visible sources for non-technical staff.
Ethics / Professional responsibility	RBAC, PII minimisation, AI disclosure, auditability and human review.

8. Conclusion

KnowledgeDesk integrates prompt engineering, RAG, AI agents and multimodal document handling into a single enterprise assistant. The calculations show that the context, storage and monthly token-cost constraints are manageable. The major deployment issue is latency: a 45-token/second model needs about 5.56 seconds for 250 output tokens. A faster generator, shorter controlled responses, caching and top-5 reranking are therefore recommended. The ZIP provides a runnable prototype with sample documents and evaluation code.

9. Student Reflection

The main lesson from integrating the four areas is that an enterprise GenAI system is more than an LLM prompt. Retrieval quality, document metadata, access control, tool routing, latency and evaluation all affect whether the answer is safe and useful. With more time, the system could be improved with a real employee-query benchmark, multilingual retrieval, stronger multimodal embeddings, distributed serving for 500 concurrent users and a real ticketing integration.

10. References

1. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.
2. Sentence Transformers documentation and all-mpnet-base-v2 model documentation.
3. FAISS documentation for dense-vector similarity search.
4. BM25 information-retrieval formulation.
5. Ollama documentation for local LLM serving.
6. Streamlit documentation for interactive AI applications.
7. PyMuPDF and python-docx documentation.
8. CSA65 assignment template supplied by the user.

11. Rubric Alignment

Criterion	Marks	Covered in
Problem understanding & formulation	10	1–2
Application of course/domain knowledge	20	2
Solution/design/implementation	20	3–4
Modern tools / techniques	10	4
Results, testing & validation	15	5
Analysis, trade-offs & justification	15	6
Broader considerations	5	7
Technical documentation & reflection	5	8–10
TOTAL	100	

Report prepared using the supplied CSA65 assignment structure and requirements.